

Math 1540 Preassignment-I

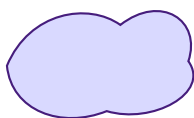
Warm-Up: What Is Integration?

Do this before class — about 15 minutes, completion credit. No printer? Just write your answers, numbered, on any sheet of paper.

Why we're doing this: You're about to spend a whole chapter on integration. Before we throw definitions and symbols at you, we want to know what you already think it means.

Part 1 — Take a Guess First (no googling yet!)

Look at the shape below. It's a made-up pond.



Question 1. How much space (area) does this pond take up? You don't know a formula for a shape like this — that's the point. Just guess a method.

- Write down a number (any unit — squares, tiles, whatever).
- Write 1–2 sentences explaining *how* you got that number.

Hint if you're stuck: what would you do if someone handed you a sheet of graph paper and this picture?

Part 2 — Pick At Least One

Go find out what "integration" actually means. Choose **at least one** option below — whichever sounds interesting to you. (Curious? Do more than one.)

1. **Watch a video.** *Essence of Calculus*, Chapter 8, by 3Blue1Brown (about 20 min).
3blue1brown.com/lessons/integration
2. **Play with an app.** Open this Desmos graph and drag the n slider. Watch what happens to the green rectangles as n grows.
desmos.com/calculator/cww0oi3dri
3. **Read something short.** Integration isn't just "area under a curve" — it's a way to multiply things that keep changing. A quick, friendly read.
[betterexplained.com: "Integrals as Multiplication"](https://betterexplained.com/articles/integrals-as-multiplication/)
4. **Ask an AI.** Ask Claude, ChatGPT, or similar: "What is integration, in simple terms?" Find **one** sentence in its answer you don't fully believe or understand yet — you'll use it below.

Question 2. Which one(s) did you pick? _____

Part 3 — Write It Down

Answer in your own words. A sentence or two each is plenty.

Question 3. Explain it to your roommate. Imagine you have to explain it to your roommate. They took Math 1530 but not 1540. In one sentence, with no formulas, tell them what integration is.

Question 4. What surprised you? One thing you learned that you didn't expect, or don't quite believe yet.

Question 5. What do you still want to know? One question about integration you want answered by the end of this chapter. We'll come back to these in class.

Question 6. How sure are you? Circle a number, 1–5.

① ② ③ ④ ⑤

just guessing

pretty confident

Turning it in: Have your answers ready before our next class. Graded on completion only.

A few of your answers to Questions 1 and 3 will show up on the board (name off) on the first day we talk about integration. Don't worry — being "wrong" here is exactly how we start.